A Harm Variance and Attacker–Target Heatmap

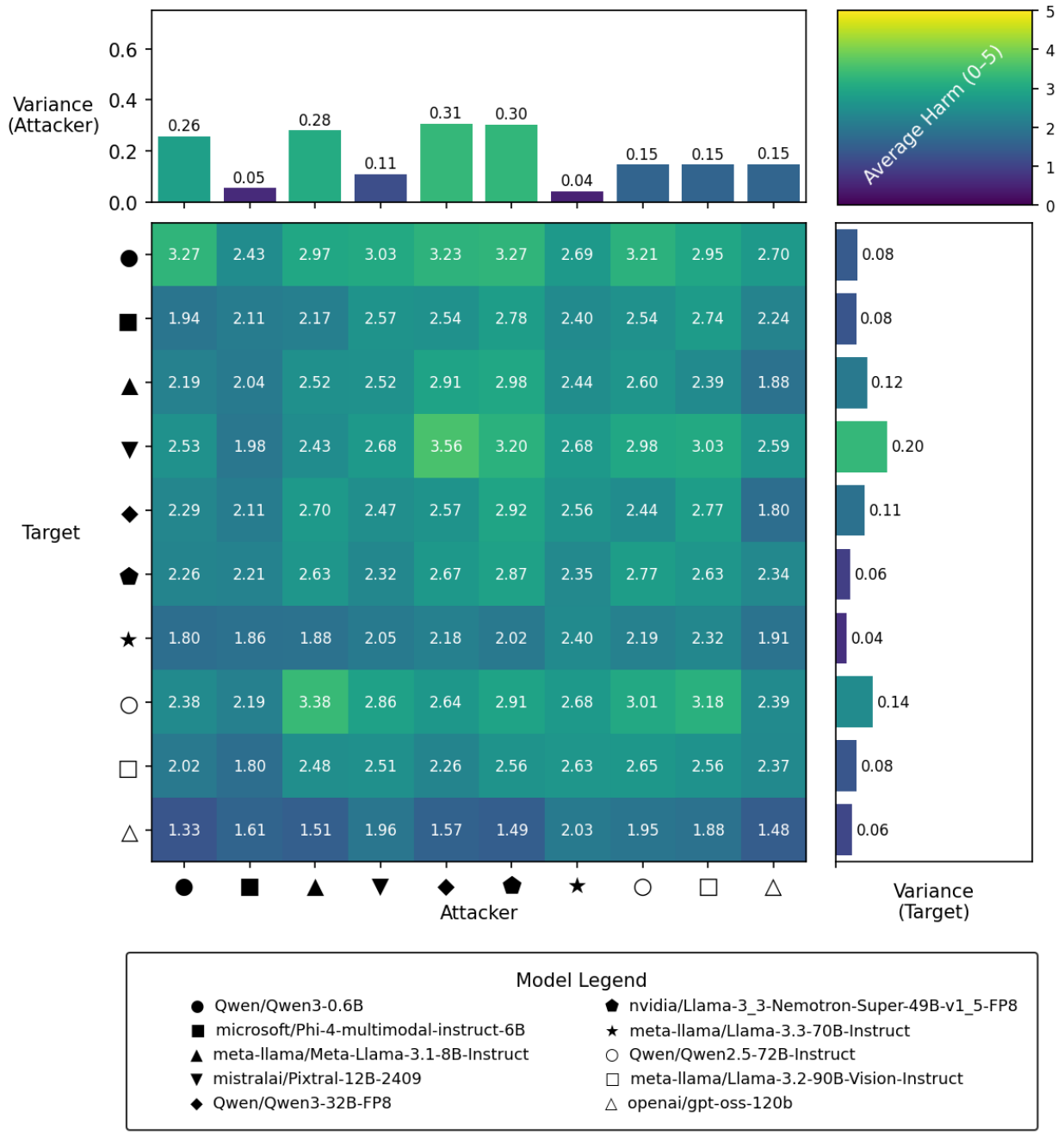

Figure 4: Combined heatmap and variance plots showing mean harm scores across attacker (columns) and target (rows) models. Bar charts indicate attacker- and target-side harm variance (0.180 and 0.097), highlighting greater dispersion among attackers than targets.

This visualization summarizes per-model differences in average harm and dispersion. It complements the main text by showing that attacker-side variability exceeds target-side variability, consistent with the variance metrics reported in Section 4.

Distribution Statement A. Approved for public release: distribution is unlimited.

B Run Scatter Matrix

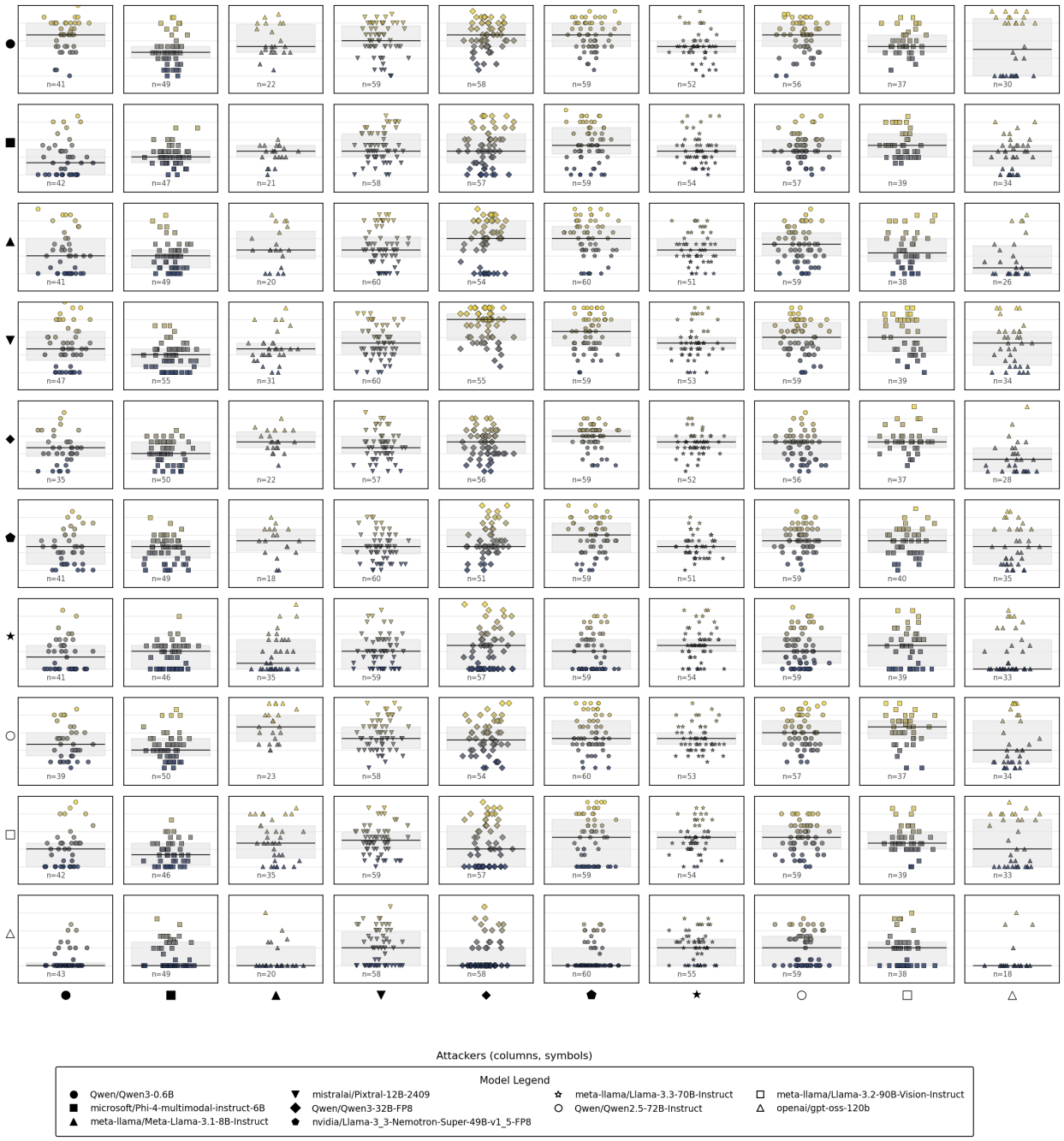

Figure 5: Scatter matrix of harm distributions across all attacker–target pairs (attacker refusals excluded). Each cell shows individual run outcomes (harm 1–5) for a given pairing, with median lines and interquartile bands summarizing variability. This visualization highlights cross-family consistency and the broader scaling pattern linking attacker size to harm intensity.

C Robustness Analysis

This appendix evaluates the robustness of the observed relationship between attacker–target size ratio and harm across architectures, prompt types, and experimental domains. We combine correlation-based and model-based approaches to verify that the main scaling trends hold independently of any single family or dataset artifact.

C.1. Leave-One-Family-Out Correlation Robustness

To assess whether the scaling relationship between attacker–target size ratio and harm was driven by a particular model family, we performed a leave-one-family-out robustness analysis. For each major architecture (*Llama*, *Qwen*, *Pixtral*, *Nemotron*, and *GPT*), all corresponding attacker–target pairs were excluded, and Pearson and Spearman correlations were recomputed between mean harm and the logarithm of the attacker-to-target size ratio.

The resulting coefficients remained stable across exclusions (Pearson $r = 0.486 - 0.562$; Spearman $\rho = 0.493 - 0.568$), indicating that the positive scaling trend persists independently of any single architecture or alignment lineage. This consistency suggests that the correlation between model size ratio and harm reflects a general cross-family pattern rather than an artifact of a specific model series or tuning pipeline.

Table 1: Leave-one-family-out correlation results. Correlations remain consistent when each major model family is excluded in turn.

Excluded Family	**Pearson** r	**Spearman** ρ
GPT	0.562	0.568
Llama	0.525	0.537
Other	0.486	0.493
Pixtral	0.507	0.499
Qwen	0.499	0.498

C.2. Mixed-Effects Model Robustness

To test which experimental factors most strongly predicted harm dispersion while controlling for prompt-level variation, we fit a linear mixed-effects model with random intercepts by prompt ID. Fixed effects included the logarithm of the attacker-to-target size ratio, attacker and target model families, the adversarial prompt variant, and the harm domain. Approximate partial R^2 values were computed for each fixed factor to estimate their unique contribution to harm variance. Attacker family explained the largest share of variance in harm ($R^2 = 0.086$), followed by size ratio ($R^2 = 0.055$), prompt variant ($R^2 = 0.033$), and target family ($R^2 = 0.030$), while harm domain contributed negligibly ($R^2 = 0.002$). These results show that both model training and scale asymmetry are strong predictors of harm outcomes, with prompt and domain effects playing a secondary role.

C.3. Integrated Interpretation

Together, the correlation and mixed-effects robustness analyses demonstrate that the scaling relationship between attacker-to-target size ratio and harm is general across architectures and not driven by any single model family. The leave-one-family-out test confirms that the size–harm correlation persists within each family group, while the mixed-effects analysis reveals that architectural